Run Wang

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Summary

Computer Science Ph.D. student specializing in robust autonomous driving, 3D perception, and mobile/edge computing, with experience in research and systems/software engineering. Author of published or accepted papers in CVPR, BMVC, INFOCOM, and IWQoS.

Skills

Programming: C, C++, C#, Python (PyTorch), Java
Tools: Docker, Kubernetes, CARLA
Research Areas: Robust Autonomous Driving, 3D perception, and Mobile/Edge Computing

Education

Clemson University — Clemson, SC
Ph.D. Student in Computer Science — *Aug. 2025 – Present*
- Concentration: Robust Autonomous Driving, 3D Computer Vision, AI Security

Carnegie Mellon University — Pittsburgh, PA
M.S. in Electrical and Computer Engineering (Applied Advanced Program) — *Sep. 2021 – May 2023*

Rensselaer Polytechnic Institute — Troy, NY
B.S. in Computer & Systems Engineering — *Sep. 2017 – May 2021*

Selected Publications and Preprints

R. Wang *et al.*, "FlexMap: Robust HD Map Construction under Flexible Camera Configurations"
arXiv Preprint, 2026.

C. Zhou, **R. Wang** *et al.*, "FF3R: Feedforward Feature 3D Reconstruction from Unconstrained Views"
CVPR, 2026.

Q. Yang, C. Zhou, X. Liu, **R. Wang** *et al.*, "Sat3R: Satellite DSM Reconstruction via RPC-Aware Depth Fine-tuning"
CVPRW, 2026.

R. Wang *et al.*, "Lightweight Detection of Abnormal Battery Drain Induced by Network Operations of Mobile Apps"
IEEE INFOCOM, 2026. Acceptance Rate: 329/1740 (18.91%)

R. Wang *et al.*, "MAPP: Predictive UI View Pre-caching for Improving the Responsiveness of Mobile Apps"
IEEE/ACM IWQoS, 2025. Acceptance Rate: 81/326 (24.8%)

X. Liu, C. Zhou, **R. Wang** *et al.*, "Lang-SVG: Hierarchical Image Vectorization with Language Priors"
NeurIPS, 2026.

P. MohajerAnsari, A. Salarpour, **R. Wang** *et al.*, "GATE: Reliability-Gated Gaussian Evidence Fusion for Training-Free Test-Time Adaptation of Vision-Language Models"
British Machine Vision Conference (BMVC), 2026. Acceptance Rate: 404/1448 (27.9%)

R. Wang *et al.*, "Adversarial Prompts for Acceptance Collapse in Speculative Decoding"
arXiv Preprint, 2026.

Manuscripts currently under review

R. Wang *et al.*, "BEVIVE: Defend against Targeted Hiding Attacks on Multi-Sensor Fusion in Autonomous Driving"

C. Zhou, **R. Wang** *et al.*, "SCOPE-4D: Endoscopic 4D Geometry Foundation Models"

M. Khan, **R. Wang** *et al.*, "PT-Flow: One-Step Schrödinger-Bridge Generation by Proximal Tilting"

Y. Liu, **R. Wang** *et al.*, "Who Touched the Screen? Driver-Passenger Identification in In-Vehicle Infotainment Systems"

Research Experience

Graduate Research Assistant Aug. 2025 – Present
Clemson University *Clemson, SC*

- Supervisors: Dr. Siyu Huang and Dr. Mert D. Pesé
- Ongoing Project: Researching AI security and robustness of autonomous-driving systems, focusing on adversarial vulnerabilities in perception and decision-making pipelines.

Graduate Research Associate May 2024 – Aug. 2025
The Ohio State University *Columbus, OH*

- Supervisor: Prof. Xiaorui Wang
- Designed ML and time-series models for mobile/edge performance optimization, producing two peer-reviewed papers accepted to IEEE/ACM IWQoS 2025 and IEEE INFOCOM 2026.

Research Assistant May 2022 – May 2023
CERLAB, Carnegie Mellon University *Pittsburgh, PA*

- Supervisors: Prof. Kenji Shimada and Dr. Wentai Zhang
- Adapted PSPNet, DeepLabV3+, and SketchGNN for component extraction from engineering drawings, achieving 84% overall accuracy on in-house dataset. (acknowledged by this paper: Component segmentation of engineering drawings using Graph Convolutional Networks)

Research Assistant Jan. 2022 – May 2022
CyLab, Carnegie Mellon University *Pittsburgh, PA*

- Supervisor: Dr. Cai Yang
- Collected and dynamically traced 80+ Linux malware samples, extracting 14 system-level behavioral features for malware behavior analysis.
- Built an in-house behavioral dataset from runtime traces to support downstream malware classification and anomaly detection experiments.

Research Assistant Jan. 2020 – Dec. 2020
RPI–IBM Cognitive and Immersive Systems Lab *Troy, NY*

- Supervisors: Dr. Jeff Kephart and Dr. Xiangyang Mou
- Built real-time gesture recognition system in C# and Python using Intel RealSense cameras, collecting 600+ gesture samples and extracting skeletal motion features for recognition experiments.

Industry Experience

Application Developer – Full Stack May 2023 – Apr. 2024
Computer Packages Inc. *Rockville, MD*

- Developed and maintained C#/.NET, JavaScript, and SQL Server features for legacy-to-modern application migration, preserving data integrity across production workflows.
- Optimized SQL Server query performance using stored procedures, indexing, and execution-plan analysis, reducing query latency by $> 50\%$.

Project Experience

Cosmospapers Research Assistant Platform Jan. 2026 – Present
Demo - Github

- Built a RAG-based research discovery platform that indexes 54,000+ papers from 15+ top CS conferences, combining vector retrieval with interactive topic exploration.
- Extracted 150 hierarchical research topics and visualized their relationships, so users can browse papers by research area instead of keyword search alone.
- Released the platform as a public demo and open-source GitHub repository.

VANTAGE Benchmark (Nvidia & Clemson University) Oct. 2026 – Present
Demo - Github

- Built a question-generation pipeline on RealSense RGB-D clips, using SAM 3 masks and sensor depth to create numeric distance, direction, and depth-difference questions
- Removed answer leaks with mirrored and time-reversed variants and balanced Yes/No pairs; scored with MRA and group-wise accuracy, and built an evaluation pipeline on Palmetto A100 nodes.